

Your genetic variants can impact your body's hormone production. Please see the linked articles for details and complete references.

[Article: Uterine Fibroids](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
ACE	rs4343	AG	G	0.49	Increased risk of fibroids
XPC	rs2228000	GG	A	0.24	Decreased risk of fibroids
XPG	rs873601	AG	A	0.68	AA: Increased risk of fibroids
CYP1B1	rs1056827	--	A	0.29	increased risk of uterine fibroids
ESR1	rs2234693	CC	T	0.53	TT: Decreased risk of fibroids
CCND1	rs9344	AG	A	0.44	Increased risk of fibroids
FASN	rs4247357	--	T	0.43	Increased risk of fibroids
GSTM1	rs366631	--	A	0.78	AA: Increased risk of fibroids

[Article: PMS / PMDD](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
HTR1A	rs6295	GG	C	0.52	Increased risk of PMDD
ESR1	rs9340799	GG	G	0.33	GG only: Increased risk of PMS
COMT	rs4680	AG	A	0.48	A/A and A/G: lower (or normal) risk of PMS; G/G: fast COMT and increased risk of PMS
BDNF	rs6265	CC	T	0.18	T/T: decreased BDNF, increase menstrual pain, altered brain activation in luteal phase in PMDD

[Article: Estrogen](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP1A1	rs1048943	TT	C	0.04	Decreased CYP1A1; Increased risk of estrogen related problems
CYP1B1	rs1056827	--	A	0.29	Increased risk of estrogen related problems
CYP3A4	rs2740574	TT	C	0.11	Increased CYP3A4; Increased estrogen metabolism to 16a-OHE1
COMT	rs4680	AG	A	0.48	AA only: low COMT; Increased risk of estrogen quinone metabolites
GSTP1	rs1695	AA	G	0.32	Reduced GSTP1; Decreased metabolism of estrogen quinone metabolites
GSTM1	rs366631	--	A	0.78	AA only: GSTM1 null; increase risk of estrogen related problems
UGT1A6	rs2070959	AA	G	0.32	Lower enzyme activity; Increased risk of estrogen problems
NQO1	rs1800566	AG	A	0.2	Low NQO1 activity; Increased risk of estrogen related problems
CYP19A1	rs4646	CA	A	0.27	AA only: lower estrogen levels
CYP19A1	rs700518	TT	T	0.53	TT only: higher estrogen levels; Increased risk of prostate problems
GP1R	rs11544331	CC	T	0.23	Decreased receptor activation; lower risk of fibroids
CYP17A1	rs743572	GG	G	0.39	Decreased risk of breast cancer
CYP1B1	rs1056836	GG	G	0.57	Decreased estradiol metabolism, increased hot flashes in smokers
HSD17B1	rs676387	--	A	0.25	A/A: slightly increased risk of breast cancer, conversion to E2
CYP1B1	rs1800440	CT	C	0.17	decreased risk of endometrial cancer; women who never used hormone therapy had a decreased breast cancer risk, while women who did use hormone therapy were at an increased relative risk
CRACD	rs4865075	GG	A	0.25	lower risk of breast cancer with hormone replacement therapy
CD80	rs7648642	--	C	0.46	higher relative risk of lobular breast cancer with hormone replacement therapy
CYP24A1	rs964293	--	A	0.38	reduced relative risk of colon cancer with estrogen plus progesterone HRT
LINC00607	rs6716449	--	C	0.06	likely increased relative risk of breast cancer with hormone replacement therapy
CYP19A1	rs12591359	--	A	0.41	slightly decreased estradiol levels (men and women)
CYP19A1	rs727479	CC	C	0.31	slightly decreased estradiol levels (postmenopausal women)
SLCO1B1	rs4149013	--	G	0.05	G/G: 4x increased relative risk of breast cancer with HRT (combined estrogen and progestin)
SLCO1B1	rs4149056	TT	C	0.14	SLCO1B1*5; higher levels of estrone sulfate, higher estrone sulfate with HRT

[Article: Testosterone](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
SHBG	rs12150660	GG	G	0.78	Lower free testosterone
SHBG	rs6258	CC	T	0.006	Lower free testosterone
SHBG	rs6259	--	A	0.1	Higher SHBG levels
SHBG	rs1799941	--	A	0.23	Higher SHBG levels
FAM9B	rs5934505	TT	T	0.72	Lower free testosterone
LIN28B	rs7759938	TC	C	0.32	Lower free testosterone
FSHB	rs10835638	GG	T	0.06	Lower free testosterone
SHBG	rs727428	TC	T	0.43	lower SHBG levels, higher DHT levels

[Article: Thyroid Levels](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TSHR	rs1991517	--	G	0.08	Increased risk hypothyroidism
TSHR	rs121908866	--	A	0.0002	Congenital hypothyroidism
PDE8B	rs4704397	AA	A	0.41	Increased serum TSH
PDE8B	rs6885099	GG	A	0.53	Decreased TSH
FOXE1	rs7850258	--	A	0.32	AA only: Decreased odds of hypothyroidism
FOXE1	rs965513	GG	A	0.31	Decreased TSH
TSHR	rs3783938	--	T	0.08	Increased risk of Hashimoto's
TSHR	rs12101255	--	T	0.37	Increased risk of Graves' (common)
TSHR	rs179247	--	A	0.53	AA: increased risk of Graves' disease (common)
TPO	rs2071403	GG	G	0.52	GG only: Increased risk of Hashimoto's and Graves'
PTPN22	rs2476601	GG	A	0.08	Slightly increased risk of autoimmune thyroid disease
DIO1	rs2235544	CA	A	0.49	AA: Decreased fT3 AC: Intermediate CC: more active DIO1, increased fT3
DIO1	rs11206244	CT	T	0.33	Higher rT3; lower fT3
DIO2	rs225014	CT	C	0.37	Decreased T4 to T3 conversion
SERPINA7	rs28933689	--	T	0	Thyroxine-binding globulin deficiency
SERPINA7	rs2234036	CC	T	0.0002	Thyroxine-binding globulin deficiency
THRB	rs28933408	--	T	0	rare mutation, thyroid hormone resistance
NR1H2	rs2695121	--	T	0.39	increased susceptibility to obesity (thyroid hormone receptor in liver that regulates T3 levels)
NR1H2	rs17373080	--	G	0.3	decreased transcription activity; increased risk of obesity (thyroid hormone receptor in liver that regulates T3 levels)
THRB	rs121918690	--	T	0	rare mutation, thyroid hormone resistance
THRB	i5003332	--	T	0	rare mutation, thyroid hormone resistance
SH2B3	rs3184504	CC	T	0.45	slightly increased risk of hypothyroidism (Hashimoto's)

[Article: Progesterone](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
PGR	rs1042838	CA	A	0.15	increased PGR, increased relative risk of ovarian cancer, uterine fibroids, breast cancer, endometrial cancer
PGR	rs471767	AA	G	0.31	linked to an increased risk of preterm birth (which can be due to lower progesterone during pregnancy)
PGR	rs1042839	--	G	0.85	decreased risk of ovarian cancer
CYP3A4	rs4987161	--	G	0.00006	CYP3A4*17, decreased function of enzyme that breaks down progesterone
CYP3A4	rs4986909	--	A	0.0001	CYP3A4*13, decreased function of enzyme that breaks down progesterone
CYP3A4	rs2740574	TT	C	0.11	CYP3A4*1B, altered function of enzyme that breaks down progesterone
CYP3A4	rs4986910	--	G	0.006	CYP3A4*3, decreased function of enzyme that breaks down progesterone
CYP3A4	rs4986907	CC	T	0.002	CYP3A4*15A, decreased function of enzyme that breaks down progesterone
CYP2C19	rs4244285	--	A	0.14	decreased metabolism of supplemental progesterone
PGR	rs10895068	--	T	0.04	possibly increased relative risk of breast cancer (studies show mixed results), endometrial cancer

[Article: CYP17A1](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
CYP17A1	rs743572	GG	G	0.39	higher CYP17A1 function; increased risk of PCOS, prostate cancer, acne; decreased breast cancer risk
CYP17A1	rs4919687	--	A	0.28	increased risk of coronary heart disease
CYP17A1	rs6162	AA	G	0.58	increased relative risk of higher PSA levels
CYP17A1	rs11191548	CC	C	0.09	decreased risk of high blood pressure
CYP17A1	rs104894154	--	T	0.00008	CYP17A1 deficiency mutation (rare)
CYP17A1	rs104894153	--	T	0.0001	CYP17A1 deficiency mutation (rare)
CYP17A1	i5001478	--	T	0.0001	CYP17A1 deficiency mutation (rare)
CYP17A1	i5001486	--	A	0.000001	CYP17A1 deficiency mutation (rare)
CYP17A1	rs104894148	--	A	0.000001	CYP17A1 deficiency mutation (rare)
CYP17A1	rs61754278	--	T	0.00007	CYP17A1 deficiency mutation (rare)
CYP17A1	i5001477	--	T	0.00008	CYP17A1 deficiency mutation (rare)

[Article: Hot Flashes](#)

Gene	RS ID	Your Genotype	Effect Allele	Effect Allele Frequency	Notes About Effect Allele
TACR3	rs77322567	CC	A	0.05	decreased risk of hot flashes
CYP1B1	rs1800440	CT	C	0.17	C/C: 3-fold greater odds of experiencing hot flashes for more than a year
AHR	rs2066853	GG	A	0.12	A/A: increased risk of hot flashes
COMT	rs4680	AG	A	0.48	A/A: in current smokers, a 13-fold increased risk of hot flashes
SULT1A1	rs1801030	--	C	0.007	SULT1A1*3 , decreased risk of hot flashes